

GENG5505 Exam Answer Guide

Section A: Ocean Reef Marina (ORM) Redevelopment case | Section B & C: Flagged topics

SECTION A — Project Lifecycle & Sustainability

Q1: The Four Stages of the Project Life Cycle and Their Contribution to Project Success

The Project Life Cycle (PLC) guides an infrastructure project from initial idea to operational reality through four sequential stages: **Concept, Planning, Execution (with Monitoring & Controlling), and Finalisation**.

Stage 1: Concept Stage (ORM: 1970s – 2016)

What it is: The initial stage where the project is conceived, justified, investigated, assessed, and approved at a high level. It captures, documents, and clarifies the **four input constraints**: Time, Cost, Resources, and Specification (Hartley 2018).

Why it is important: One of the most common causes of project failure is miscommunication and misalignment between stakeholder expectations and the final output. The Concept Stage enforces the fundamental principle: *identify the problem before attempting to solve it*. Without it, the project lacks a justified purpose, clear boundaries, or preliminary approval.

Application to ORM: The Concept Stage spanned nearly four decades (1970s–2016) under the City of Joondalup's leadership. The original concept was to transform a small boat harbour into a world-class marine precinct. However, a critical governance gap existed: the City held high *interest* but lacked the statutory *power* and capital to navigate the environmental complexities of the adjacent Marmion Marine Park. Despite early land acquisition efforts from 1979, the project stalled without formal justification or approved scope. The turning point came when the State Government appointed DevelopmentWA as lead proponent in 2016 — establishing authority, intent, and strategic justification at the appropriate governance level. This demonstrates that a weak Concept Stage leads to decades of planning inertia.

Stage 2: Planning Stage (ORM: 2016 – 2020)

What it is: All work required for successful project completion is planned and scheduled. Objectives are finalised, resources allocated, quality standards agreed, final costs approved, timing confirmed, and risks formally identified.

Why it is important: Without a detailed plan, execution lacks direction, resource alignment, and budget control. The planning stage prevents costly surprises and consolidates stakeholder alignment — misalignment caught here is far cheaper to address than during construction.

Application to ORM: The Planning Stage was defined by DevelopmentWA's formalisation of the project's statutory and technical framework. The most significant planning decision was implementing **Improvement**

Plan No. 51 (IP 51) under the *Planning and Development Act 2005*, granting the WAPC authority to bypass local planning schemes, thereby reducing susceptibility to political interference. The State Government committed **\$120 million** for environmental approvals, detailed design, and forward infrastructure works. Project scope was formally defined across four precincts: Waterways, Mixed-Use/Waterfront, Residential, and Marine Enterprise. However, centralised decision-making reduced community consultation, drawing pushback from environmental groups regarding marine habitat modelling — a residual social licence risk.

Stage 3: Execution Stage (ORM: 2021 – Present)

What it is: The planned work is physically implemented. Emphasis shifts to **monitoring and controlling** — tracking actual progress against Stage 2 schedules, managing change, resolving issues, and maintaining stakeholder communication.

Why it is important: Most project expenditure occurs here; poor execution directly causes cost overruns, delays, and quality failures. Monitoring and control mechanisms allow early detection and correction of deviations before they compound.

Application to ORM: Execution focused on constructing major breakwaters, land reclamation (~61.9 hectares), and protecting the Water Corporation's Beenyup ocean outfall pipe — a critical technical constraint. A **Preventative Control Strategy (ISO 31000)** was adopted, embedding pipe protection into the breakwater design rather than treating it reactively. The pursuit of an **ISCA Gold Rating** institutionalised environmental performance into the construction contract itself (e.g., use of recycled materials for breakwaters), ensuring the Triple Bottom Line was a contractual obligation rather than an aspiration.

Stage 4: Finalisation Stage (ORM: Future)

What it is: The completed project is formally handed over to the client. Key activities include acceptance testing, client sign-off, lessons learned documentation, resource reallocation, and post-implementation review.

Why it is important: Project success cannot be measured solely by whether technical scope was delivered. The finalisation stage formally evaluates whether the project achieved its strategic purpose and stakeholder value. Formal handover establishes clear accountability for asset ownership.

Application to ORM: The Finalisation Stage represents the transition from "Construction Project" to "Functional Precinct." Success must be measured against specific socio-economic targets — including 8,600 construction jobs and \$3 billion in economic activity. A formal **Benefits Realisation Management (BRM)** framework is essential to prevent "strategic drift," particularly given split asset ownership (marine infrastructure → Department of Transport; land assets → City of Joondalup). The finalisation phase must also incorporate long-term environmental monitoring aligned with ISO 14001 principles — transitioning from construction compliance to adaptive management. ORM represents a rare opportunity for a Post-Implementation Review capturing institutional knowledge about why the project stalled 40 years and succeeded in 5.

Q2: Sustainable Approach, Triple Bottom Line (TBL), and Application Across All Four Stages

Defining a Sustainable Approach

A **sustainable approach** is an organisation's ability to achieve its business goals and increase long-term stakeholder value by integrating economic, environmental, and social opportunities into its business strategies (Robbins et al. 2009).

Sustainability is fundamentally defined as: *meeting the needs of people today without compromising the ability of future generations to meet their own needs.*

The key insight: the long-term financial interests of a company are **not** mutually exclusive with acting fairly in the interests of stakeholders — they are mutually reinforcing through a TBL framework.

The Triple Bottom Line (TBL) Framework

Dimension	Guiding Question
Prosperity (Economic)	Will the project generate lasting prosperity and enhance affected economies?
People (Social)	Will the project be implemented in a socially responsible manner and benefit affected communities fairly and equitably?
Planet (Environmental)	Will the project cause long-term damage to the environment?

TBL Applied Across the Four Stages of ORM

Stage 1: Concept Stage — Establishing Sustainability as a Foundation

Integrating TBL from the beginning means sustainability is **built into the project's justification**, not added as an afterthought.

- **Economic:** ORM was explicitly framed as an economic catalyst — generating employment in retail, tourism, and marine services. This formed the core justification for State Government investment.
- **Social:** Community benefit was incorporated from the outset — public open spaces, a safe internal swimming beach, upgraded recreational facilities, and diverse housing options. Cultural heritage obligations were acknowledged through recognition of the Whadjuk Noongar people's connection to the coastline.
- **Environmental:** Proximity to the Marmion Marine Park and Bush Forever Site 325 meant environmental risks were identified as fundamental constraints during conceptualisation. Failing to account for these early would have invalidated the project.

Why it improves the project: Identifying TBL constraints in the concept stage prevents the costly scenario where a project is approved on economic grounds alone, then halted by environmental or social objections.

Stage 2: Planning Stage — Embedding TBL into Governance and Design

The planning stage translates TBL commitments into **formal plans, risk registers, approvals, and contractual obligations.**

- **Economic:** The \$120 million State Government allocation was formally structured to fund environmental approvals and detailed design, grounding economic promises in a realistic funding plan.
- **Social:** Stakeholder consultation was formalised through a Government Steering Committee and MOU between the State and City of Joondalup. Mendelow's Matrix was applied to map community groups and define appropriate engagement strategies. However, the top-down IP model reduced community inclusion, straining social licence.
- **Environmental:** Compliance with **Ministerial Statement 1107** — covering water quality, coastal process management, and marine habitat protection — was embedded into the approvals framework. Protection of the Beenyup ocean outfall pipe was addressed through preventative design.

Why it improves the project: Incorporating TBL into planning ensures environmental approvals, stakeholder agreements, and sustainability targets are binding commitments, reducing costly compliance failures or community conflict during construction.

Stage 3: Execution Stage — Operationalising TBL Through Delivery

During execution, TBL becomes an **active management tool** governing day-to-day construction decisions.

- **Economic:** The adaptive management approach — addressing Beenyup pipe protection through design rather than reactive intervention — prevented cost escalation. The ISCA framework also protected long-term economic returns by preserving social licence.
- **Social:** Construction affected local amenity through noise, dust, and restricted beach access. Monitoring against Ministerial Statement 1107 maintained community trust. ISCA also included local procurement and workforce development targets.
- **Environmental:** The **ISCA Gold Rating** pursuit institutionalised environmental performance: recycled materials for breakwater construction, energy-efficient lighting, water-sensitive urban design, and multiple measures protecting marine habitat, terrestrial vegetation, and the local abalone fishery.

Why it improves the project: Sustainable execution reduces waste of labour, materials, and energy while generating intellectual property and reputation value yielding long-term stakeholder returns.

Stage 4: Finalisation Stage — Evaluating Holistic Project Success Against TBL Targets

- **Economic:** BRM framework to pursue a 3:1 return on the \$120M State investment through precinct activation and commercial curation, not merely assumed.
- **Social:** Project success depends on whether the precinct delivers genuine community benefit: accessible public spaces, operational marine clubs, vibrant retail, diverse housing, and ongoing cultural heritage monitoring for the Whadjuk Noongar community.
- **Environmental:** Transition from construction-phase compliance to long-term **adaptive environmental management** under ISO 14001, with decades-long monitoring of water quality and seagrass health in the Marmion Marine Park.

Why it improves the project: Applying TBL as a quality assurance mechanism at finalisation ensures the project is not declared successful on physical completion alone. A TBL review also builds positive public image, demonstrates responsible governance, and benchmarks future state-led infrastructure projects.

Summary Table

Stage	Economic (Prosperity)	Social (People)	Environmental (Planet)
Concept	Economic catalyst framing; state funding rationale	Community benefit (beach, housing, recreation); Noongar acknowledgement	Marmion Marine Park and Bush Forever constraints identified
Planning	\$120M budget structure; IP 51 reduces approval risk	MOU with City of Joondalup; stakeholder mapping (Mendelow's)	Ministerial Statement 1107; outfall pipe risk treated by design
Execution	Adaptive risk management prevents cost blowouts; ISCA supports reputation	Community amenity monitoring; local procurement under ISCA	ISCA Gold Rating; recycled materials; habitat and fishery protection
Finalisation	BRM framework for 3:1 ROI; precinct activation	Community outreach; cultural heritage monitoring; public amenity	ISO 14001 long-term monitoring; adaptive seagrass/water quality management

SECTION B — Project Management (Flagged Topic Questions)

B1: Numerical vs Non-Numerical Methods of Project Categorisation

Non-numerical methods select projects based on qualitative criteria — strategic alignment, necessity, or managerial judgment — without financial calculation.

Examples (WK2a):

- **Sacred Cow** — A project personally championed by a senior executive or influential person. It proceeds because of who wants it, regardless of financial merit. *Example: a CEO insisting on a flagship office redesign.*
- **Operating Necessity** — A project required to maintain current operations or respond to a crisis. *Example: repairing a flood-damaged server room so the business can function.* Not optional; it must be done.
- **Competitive Necessity** — A project needed to maintain competitive advantage or match competitor offerings. *Example: launching a mobile app because all competitors already have one.*
- **Product Line Extension** — Extending an existing product or service range. *Example: adding an electric model to a car manufacturer's lineup.*
- **Comparative Benefit** — Used when an organisation has multiple projects with differing benefits and must prioritise through qualitative comparison. *Example: a council comparing park upgrades across suburbs.*

Numerical methods use financial metrics to compare and rank projects objectively:

1. **Payback Period** — Time to recover the initial investment. Formula: $\text{Cost} \div \text{Annual cash revenue}$. Simple but ignores cash flow after recovery and the time value of money.
2. **Return on Investment (ROI)** — Overall profit as a percentage of total investment. Formula: $(\text{Net profit} \div \text{Total investment}) \times 100$. Considers the entire cash-flow period but **does not account for time value of money**.
3. **Net Present Value (NPV)** — **Most exhaustive**. Projects profitability by discounting future cash flows at a stated interest rate. Formula: $\text{NPV} = \sum (\text{Cash flow} \times \text{Discount factor})$, where $\text{Discount factor} = 1 / (1 + i)^n$. Accounts for both the entire cash-flow period **and** the time value of money.

Which is most exhaustive? NPV — it is the only method that accounts for both the full timeline of cash flows and the time value of money.

B2: Five Levels of Project Management Maturity

Project management maturity models (e.g. PMI's OPM3, Kerzner's PMMM) describe how organisations progress from ad hoc project delivery to sustained, optimised performance. Five levels:

Level 1 — Initial / Ad Hoc

- No formal project management processes exist.
- Projects are run based on individual heroics; success depends on the person, not the process.
- Outcomes are unpredictable; cost and time overruns are common.
- *Characteristic:* "We do whatever it takes to get it done."

Level 2 — Repeatable

- Basic processes are established and repeated across similar projects.
- Basic planning, scheduling, and cost tracking are used.
- Success is project-dependent and not consistently transferable.
- *Characteristic:* Past successes can be repeated but processes are not yet documented.

Level 3 — Defined

- Standardised, documented project management processes are defined organisation-wide.
- A common methodology (e.g. PMBOK-aligned) is understood and applied consistently.
- Processes are integrated: scope, time, cost, quality, risk, and communication are all managed formally.
- *Characteristic:* "We have a process and everyone follows it."

Level 4 — Managed (Quantitatively)

- Project and process performance is measured quantitatively (e.g. via EVM, KPIs, dashboards).
- Data is collected and used to predict and control outcomes.
- Variation in project performance is understood and managed.
- *Characteristic:* "We measure what we do and use data to manage it."

Level 5 — Optimising

- Continuous improvement is institutionalised.
- Data from Level 4 drives ongoing refinement of processes.
- Focus shifts to innovation, preventing defects, and optimising performance.

- *Characteristic:* "We continuously improve our way of managing projects."

Progression from Level 1 to Level 5: An organisation progresses by first acknowledging the need for structure (L1→L2), then formalising and standardising processes (L2→L3), then measuring performance (L3→L4), and finally embedding a culture of continuous improvement driven by data (L4→L5). Executive sponsorship, training investment, and a willingness to learn from failure are essential at each transition.

B3: Sacred Cow vs Operational Necessity

Both are **non-numerical** project selection methods, but their drivers are entirely different.

	Sacred Cow	Operational Necessity
Driver	Senior executive preference or political will	Operational or crisis-driven requirement
Justification	"The boss wants it" — little formal financial analysis required	"We must do this or operations fail"
Risk if refused	Political or career consequences	Business continuity failure
Example	A CEO mandates a new corporate headquarters redesign to project status	A factory replaces flood-damaged machinery to resume production
Financial scrutiny	Minimal — approval precedes analysis	Minimal — necessity overrides ROI calculation
Ethical concern	Resources may be allocated inefficiently; legitimate projects displaced	Genuine necessity — generally uncontroversial

Key distinction: Sacred Cow projects are driven by *power and influence*; Operating Necessity projects are driven by *genuine operational need*. A Sacred Cow can be wasteful if the sponsor lacks good judgment; an Operating Necessity is almost always justified but should still be costed efficiently.

B4: Key Stakeholders + Why Stakeholder Identification is Important

Definition: A stakeholder is any person, group, or organisation that can affect or be affected by the project's outcome.

Key stakeholder groups (WK2b):

1. **Client** — The person/entity wanting the project to exist; ensures alignment with strategic needs; will not want an immoral or negative outcome.
2. **Sponsor** — Has ultimate accountability and responsibility; obtains funding; monitors costs/benefits/risks; has the final word on whether the project is carried out.
3. **Project Steering Group** — Senior management team responsible for adoption and uniform application of the PM framework; ensures everyone — including minorities — has a voice; approves commencement and sign-off at each phase.
4. **Project Manager** — Appointed to manage the project; leads by example; ensures ethical team performance; protects whistle-blowers.

5. **Project Team Members** — Subject-matter experts performing all required work; complete tasks ethically; communicate emergent issues honestly.

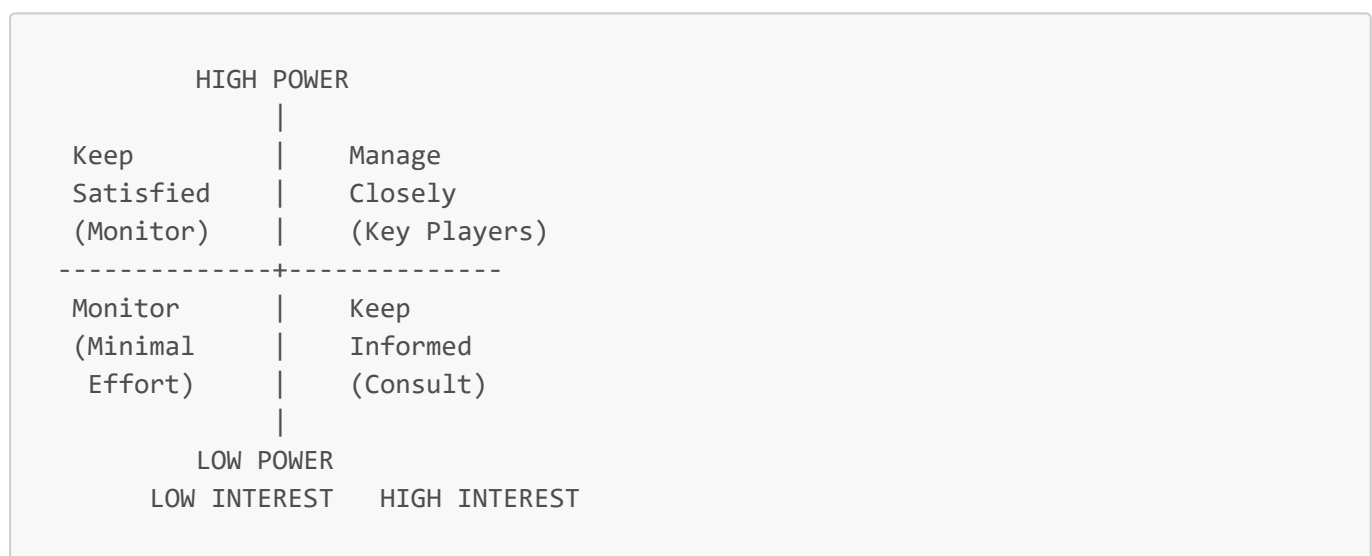
External stakeholders also include: regulators, community groups, contractors, suppliers, and the media.

Why stakeholder identification is important (WK2b):

- Stakeholders **will** impact ultimate success — they cannot be ignored.
- They must be identified prior to commencement and throughout the project.
- Their information needs must be managed throughout the lifecycle.
- They bring diversity of ideas, contributions, support, and challenges.
- Poor stakeholder management risks projects going over time, over budget, and failing to meet targets.
- Good stakeholder management also benefits future business — recommendations and re-hire opportunities.

B5: Power–Interest Matrix (Mendelow's Matrix)

The **Power–Interest Matrix** (Mendelow 1991) plots stakeholders on two axes — **Power** (ability to influence the project) and **Interest** (degree of concern about the project's outcome) — to determine appropriate engagement strategies.



Four quadrants:

Quadrant	Power	Interest	Strategy
Manage Closely	High	High	Key players — engage fully, involve in decisions, communicate frequently and in detail
Keep Satisfied	High	Low	Keep them satisfied; consult on major decisions; avoid alienating them
Keep Informed	Low	High	Communicate regularly; consult on issues affecting them; listen to their concerns
Monitor	Low	Low	Minimal effort; monitor for changes in their position

Application to ORM: DevelopmentWA (High Power/High Interest) → Manage Closely. Environmental groups (Low Power/High Interest) → Keep Informed (though this was under-managed, creating social licence problems). The City of Joondalup (High Power/High Interest) → Manage Closely via the formal MOU.

B6: Power–Interest Matrix vs RACI Matrix vs PARIS Matrix

These three tools serve different but complementary purposes in stakeholder and team management.

	Power–Interest Matrix	RACI Matrix	PARIS Matrix
Purpose	Classify stakeholders by influence and interest to prioritise engagement	Assign responsibility for project tasks/decisions	Define roles for project activities
What it maps	Stakeholders vs Power & Interest axes	People/Roles vs Tasks/Deliverables	People/Roles vs Tasks
Letters/Quadrants	4 quadrants (Monitor, Keep Satisfied, Keep Informed, Manage Closely)	R esponsible, A ccountable, C onsulted, I nformed	P articipant, A ccountable, R everview required, I nput required, S ign-off required
When to use	Early stages — to determine how much energy to invest in each stakeholder	Planning stage — to ensure clear task ownership and prevent duplication	Planning stage — where more granular role definition is needed, particularly for sign-off and review
Output	Engagement strategy per stakeholder group	Clear responsibility assignments per task	Detailed role-clarity for complex or highly governed projects

Key distinction: The Power–Interest Matrix is an *analytical* tool for external stakeholder management; RACI and PARIS are *operational* tools for internal team role clarity. PARIS is generally more detailed than RACI, with the addition of "Sign-off" and "Review required" roles that better suit regulated or complex project environments.

B7: Communication Tailored to Different Stakeholders

Effective project communication requires tailoring **frequency, format, and content** to the specific needs and position of each stakeholder (WK5b).

Communication frequency:

- **High Power / High Interest stakeholders** (e.g. sponsors, steering committees) → Frequent, regular updates; weekly or fortnightly formal meetings; immediate notification of significant risks or changes.

- **High Interest / Low Power stakeholders** (e.g. community groups) → Regular updates at key milestones; open channels for questions; avoid communication gaps that breed rumour.
- **Low Interest / Low Power stakeholders** → Periodic summaries; exceptions-based communication.

Communication format:

- Senior executives / sponsors → Concise executive summaries, dashboards, and formal written reports. Visual formats (charts, traffic-light status) are effective.
- Technical team members → Detailed specifications, schedules, and technical documentation.
- Community / general public → Plain language, accessible formats (newsletters, public meetings, web updates).
- Regulatory bodies → Formal written submissions, legally precise language, compliance reports.

Communication content:

- **Fact** (indisputable truth), **Fantasy** (opinion/interpretation), **Folklore** (rumour/hearsay), and **Feelings** (intuition/ego/emotion) must be distinguished (WK5b). Stakeholder communications should prioritise fact and clearly separate it from interpretation.
- Content should be **clear, concise, accurate, timely, targeted, appropriate, and complete** — delivered to the right audience, in the right format, at the right time, with the right impact.

Communication barriers to manage: lack of client involvement; poorly informed stakeholders; lack of reporting requirements; poor documentation; frequent scope change; changing personnel (WK5b).

B8: WBS — Limitations and Why It Is Insufficient for Scheduling Alone

Work Breakdown Structure (WBS): A hierarchical decomposition of the total project scope into smaller, manageable work packages. It defines *what* must be done.

Advantages of WBS:

- Provides a complete picture of all project deliverables
- Enables accurate cost and resource estimation
- Clarifies scope and prevents work from being overlooked
- Supports team accountability through clear work package ownership
- Useful basis for risk identification

Limitations of WBS (why it is insufficient for scheduling on its own):

1. **No sequencing / dependencies** — A WBS shows *what* must be done but not *in what order*. It does not capture the logical relationships between tasks (e.g. "Task B cannot start until Task A is complete").
2. **No timeline** — A WBS has no time axis. It cannot show when tasks start or finish, or the overall project duration.
3. **No critical path** — It cannot identify which sequence of tasks determines the minimum project duration.
4. **No resource allocation over time** — It lists resource requirements per work package but does not show resource loading across the schedule.
5. **No progress tracking** — It is a static decomposition; it cannot be used to monitor actual vs planned progress.

Why PERT is needed to complement WBS: PERT (Program Evaluation and Review Technique) is a network diagram that adds sequencing, dependencies, time estimates (using the 3-point formula: $T_e = (T_o + 4T_m + T_p) / 6$), and identifies the **critical path** — the longest chain of dependent tasks that determines the minimum project duration. Together, WBS defines scope and PERT defines schedule logic.

Example WBS (simplified):

Project

- ├ 1. Design
 - ├ 1.1 Concept design
 - └ 1.2 Detailed design
- ├ 2. Construction
 - ├ 2.1 Site preparation
 - └ 2.2 Building works
- └ 3. Handover

→ WBS alone cannot tell you that 2.1 must precede 2.2,
or that Design must complete before Construction begins.
PERT/network diagram adds this sequencing.

B9: Fixed Price vs Cost Reimbursable vs Time & Materials Contracts

(WK6b)

	Fixed Price	Cost Reimbursable	Time & Materials (T&M)
Definition	Delivery of a well-defined deliverable for a pre-agreed fixed price	Seller is paid for actual costs incurred plus a fee (profit)	Combines fixed hourly/daily rates (fixed-price element) with reimbursement of materials (cost element)
Best used when	Scope is well-defined and stable; risk is low	Scope is uncertain or evolving; seller expertise is critical	Scope is partially defined; short-term or supplemental work
Customer risk	At mercy of sole-source contractor; incomplete specs lead to costly variations	Final cost unknown; open-ended; rising costs if scope expands	No defined end point; open-ended exposure
Contractor risk	Careful cost/schedule estimation required; cost growth → unprofitable	Fee percentage may decline as costs rise; reputation risk	Fixed rates may not cover actual costs if conditions change
Risk allocation	Primarily on the contractor	Primarily on the client	Shared
Advantages	Budget certainty for client; incentivises contractor efficiency	Flexibility; appropriate for R&D or complex projects	Flexibility; quick to execute

	Fixed Price	Cost Reimbursable	Time & Materials (T&M)
Disadvantages	Inflexible; contractor may cut quality to protect margin	No cost certainty; requires strong oversight	No cost ceiling; requires trust
Example	Building a house to fixed specifications	Developing custom software with evolving requirements	Consulting engagement, IT support services

Legally binding contract requirements (WK6b): Offer, Acceptance, Intention/Consent, Consideration, Mutuality, Capacity, Legality — plus express and implied terms, conditions and warranties, and provision for breach and remedy.

B10: Cost Management vs Budget Management

These are related but distinct disciplines in project finance.

Cost Management is the *ongoing process* of planning, estimating, allocating, monitoring, and controlling project costs throughout the entire project lifecycle. It asks: *Are we spending in line with what we planned, and what will we have spent by the end?*

Budget Management is the *establishment and governance of the financial framework* — the approved baseline against which costs are measured. It asks: *What is our approved spending limit, and are we working within it?*

	Cost Management	Budget Management
Focus	Controlling and optimising actual expenditure	Setting and maintaining the approved financial baseline
Timing	Continuous throughout execution	Primarily set in planning; updated only through formal change control
Tools	EVM, cost tracking reports, forecasts (EAC, ETC)	Budget plan, change control process, contingency reserve
Question answered	"How much have we spent and what will we spend?"	"How much are we allowed to spend?"

Main budgeting techniques (WK4a):

- **Traditional** — Previous year's level as the foundation for next year's figures.
- **Zero-based** — Every activity justified from scratch; starts at zero.
- **Program** — Activities grouped by program or major activity.
- **Top-down** — Senior managers pool knowledge; estimates passed down for further breakdown.
- **Bottom-up** — Individuals responsible for work estimate their tasks; aggregated to total.

B11: Top-Down vs Bottom-Up Budgeting

	Top-Down Budgeting	Bottom-Up Budgeting
How it works	Senior managers set overall budget based on experience and past results; figure is allocated down to teams	Individual task owners estimate their own costs in detail; estimates are aggregated to the total
Advantages	Fast to produce; draws on executive experience; useful when detail is unavailable; provides strategic alignment	More accurate at task level; team ownership improves commitment; identifies risks early; reflects reality of the work
Disadvantages	May be unrealistic or arbitrary; teams may feel imposed upon; insufficient detail for complex projects	Time-consuming to compile; may be overly optimistic (padding); requires significant coordination; can miss strategic constraints
Best used when	Early-stage estimates; feasibility studies; when time is limited	Detailed planning stage; complex projects where task granularity matters

B12: Analogous Estimating, Expert Judgement, and Three-Point Estimating

These are three common cost (and time) estimation techniques used in project planning.

Analogous Estimating (Top-down): Uses data from similar past projects as the basis for estimating the current project. *Example: "Our last hospital project cost \$1.2B, so a similar-sized hospital should cost approximately the same."* Fast and low-effort but relies on comparability of projects. Less accurate when the new project has significant differences in scope, technology, or context.

Expert Judgement: Solicits the knowledge and experience of subject-matter experts (individuals or panels) to estimate costs or durations. Useful when historical data is unavailable or the project involves novel technology. *Example: bringing in a specialist contractor to estimate the cost of decommissioning a nuclear facility.* Strength: captures tacit knowledge. Weakness: subjective; prone to optimism bias or anchoring.

Three-Point Estimating (PERT-based): Addresses uncertainty by using three estimates for each task:

- **To** = Optimistic time (best case)
- **Tm** = Most probable time
- **Tp** = Pessimistic time (worst case)

Formula (Beta distribution, PMBOK): $Te = (To + 4Tm + Tp) / 6$

Example: To=5 days, Tm=10 days, Tp=30 days → $Te = (5 + 40 + 30) / 6 = 12.5$ days.

Produces a weighted average that accounts for uncertainty without ignoring best- and worst-case scenarios. More accurate than single-point estimates for complex or novel activities.

B13: Challenges of Estimating Project Costs + How to Overcome Them

Key challenges:

1. **Scope uncertainty** — If the scope is not fully defined, estimates are built on assumptions. *Overcome by:* completing thorough scope definition before estimating; using parametric or analogous estimates for undefined elements with explicit contingency.
2. **Optimism bias** — Estimators (and sponsors) systematically underestimate costs and duration ("planning fallacy"). *Overcome by:* three-point estimating; independent review; reference class forecasting from comparable past projects.
3. **Changing scope (scope creep)** — New requirements added during execution inflate costs beyond original estimates. *Overcome by:* rigorous change control processes; formal scope baseline; requiring sign-off for all scope changes.
4. **Lack of historical data** — Novel projects or new technologies have no precedent. *Overcome by:* expert judgement; detailed bottom-up estimation; pilot studies or prototypes.
5. **Risk and uncertainty** — Unknown unknowns cannot be priced in advance. *Overcome by:* including contingency reserves (management reserve for unknown unknowns; contingency reserve for identified risks); Monte Carlo simulation for probabilistic cost modelling.
6. **Inflation and market volatility** — For long-duration projects, material and labour costs may shift significantly. *Overcome by:* escalation clauses in contracts; phased budgeting with periodic re-baselining.

B14: Myers-Briggs Personality Assessment in Project Team Management

The **Myers-Briggs Type Indicator (MBTI)** is a personality assessment tool (based on Jung's theory) that classifies individuals along four dimensions:

- **E/I** — Extraversion vs Introversion (how energy is directed)
- **S/N** — Sensing vs iNtuition (how information is processed)
- **T/F** — Thinking vs Feeling (how decisions are made)
- **J/P** — Judging vs Perceiving (how the world is organised)

This produces 16 personality types (e.g. INTJ, ENFP).

Use in project team management (WK5a):

- Helps project managers understand *why* team members communicate and respond differently.
- Enables better role assignment — e.g. Sensing types often excel at detail-oriented tasks; Intuitive types at big-picture planning.
- Supports conflict prediction — Thinking vs Feeling clashes are common; awareness allows proactive management.
- Assists in team composition — deliberate diversity of types can strengthen team performance.
- Provides a shared vocabulary for team members to discuss working styles without personal judgment.

Limitations / criticisms:

- MBTI has been criticised for low test-retest reliability — individuals often receive different types on retesting.
- It creates binary categories (E or I) when most people fall on a spectrum.

- Risk of labelling and stereotyping — "I'm an INTJ so I don't do feelings" can excuse poor behaviour.
- Should be used as a *starting point for dialogue*, not a definitive classification system.
- Most useful when team members voluntarily engage with it; coercive use can backfire.

B15: Major Conflict Management Strategies and When to Use Each

(WK5a — Thomas-Kilmann model, assertion/cooperation axes)

Strategy	Assertion	Cooperation	Outcome	Use when...
Avoiding	Low	Low	Lose-lose	Issue is minor; confrontation causes more damage than the issue; time-out is needed; inequitable power balance
Competing / Forcing	High	Low	Win-lose	You know you are right; stakes too high to lose; quick decisive action needed; unpopular decisions must be made
Accommodating	Low	High	Lose-win	Other's evidence is more compelling; peace/goodwill is more important; want to create tactical advantage; want to avoid damaging a relationship
Collaborating	High	High	Win-win	Want to build alliance and enduring commitment; need optimal outcome without sacrificing key interests; consolidating different perspectives
Compromising	Mid	Mid	Partial win/Partial win	Outcomes are moderately important; no other option works; balance of power is even; a "moving forward" gesture is needed

If conflict is rapidly evolving: Competing / Forcing may be needed in the short term to halt escalation, with a plan to shift to **Collaborating** once the immediate crisis is stabilised. Avoiding can be used temporarily to buy time, but should not be a long-term response.

B16: Information-Gathering Techniques — Interviews, Questionnaires, Workshops, Historical Data Analysis

Used in scope definition, requirements gathering, and stakeholder analysis.

Technique	Advantages	Disadvantages
Interviews	Rich, in-depth qualitative data; can probe for clarification; builds rapport; uncovers hidden concerns	Time-consuming; interviewer bias; not scalable for large stakeholder groups; responses hard to compare

Technique	Advantages	Disadvantages
Questionnaires / Surveys	Scalable; fast; anonymous → more honest responses; easy to quantify and analyse	Low response rate; cannot probe for depth; poorly worded questions skew results; misses non-verbal cues
Workshops (Focus Groups)	Collective intelligence; builds shared understanding; stakeholder buy-in; surfaces conflicting views in one session	Dominant voices can suppress minority views; groupthink risk; requires skilled facilitation; time-intensive to organise
Historical Data Analysis	Objective; based on real outcomes; supports analogous estimating; no respondent bias	Data may be incomplete, outdated, or not comparable; past ≠ future for novel projects; relies on quality of records

Best practice: Combine methods — e.g. use historical data to inform interview questions, run workshops to validate findings, and use questionnaires to confirm with a broader population.

B17: Scope Creep — Concept, Triggers, and Control

Definition (WK3a): Scope creep is the uncontrolled expansion of a project's scope beyond its original agreed boundaries, typically without corresponding adjustments to time, cost, or resources. Other terms: innovation, continuous improvement, client change of mind, poor management.

Common triggers (WK5b):

- Poor initial requirements definition — scope was never clearly specified
- Unanswered questions on deliverables left open for "later"
- Lack of stakeholder involvement — unstated needs emerge mid-project
- "Exceeding expectations" mentality — team adds features to impress
- Discovery of new solutions during execution
- Ineffective project manager who cannot say no
- External environmental changes (regulatory, market, technology)

How to control scope creep:

- **Establish a clear scope baseline** at the end of the planning stage — a documented, agreed definition of what is included and excluded.
- **Implement formal change control** — all scope changes must go through a written approval process, assessed against all other project variables (time, cost, risk, quality, TBL).
- **Use a Scope Management Plan** that documents how scope will be defined, validated, and controlled (PMBOK).
- **Engage stakeholders early** — involve them in scope definition so their needs are captured upfront, not discovered mid-execution.
- **MoSCoW prioritisation** — classify requirements as Must, Should, Could, Won't to create a defensible priority order (see B18).
- **Regular scope reviews** — compare current project activities against the approved scope baseline at each reporting cycle.

B18: MoSCoW Prioritisation Technique for Scope Management

MoSCoW is a requirements prioritisation technique used to manage project scope by classifying each feature or requirement into one of four categories:

Category	Meaning	Implication
Must have	Non-negotiable; project fails without these	Must be delivered within the agreed timeframe and budget
Should have	Important but not critical; significant value	Delivered if at all possible; deprioritised if constraints tighten
Could have	Desirable; "nice to have"	Included only if time and budget allow; first to drop in trade-off decisions
Won't have (this time)	Explicitly out of scope for this iteration	Acknowledged but deferred to future phases; prevents scope creep by naming exclusions

How it helps manage scope: MoSCoW creates a transparent, agreed priority order among stakeholders. When scope creep pressure arises (a stakeholder requests an addition), the team can objectively assess where it falls in the hierarchy. If it is a "Could have," it can be formally declined or deferred without conflict. If it is genuinely a "Must have," it triggers formal change control with corresponding time/cost adjustments.

Example: In a hospital IT system implementation, "Must have" = patient record access; "Should have" = automated appointment reminders; "Could have" = patient-facing portal; "Won't have (this phase)" = AI diagnostic tools.

B19: Gantt Chart — Purpose, Advantages, and Disadvantages

Purpose: A Gantt Chart is a horizontal bar chart that provides a visual timeline of project activities, showing their start dates, end dates, durations, and overlap. It is one of the most widely used project scheduling tools.

Advantages:

- Clear, intuitive visual representation of the project schedule
- Easy to communicate to stakeholders of all technical levels
- Shows task durations and overlap (parallel vs series activities) at a glance
- Useful for monitoring progress — actual vs planned bars can be overlaid
- Supports resource planning by showing when activities occur
- Can incorporate milestones and dependencies

Disadvantages:

- Becomes difficult to read on complex projects with many tasks
- Does not clearly show task dependencies or the critical path (unlike PERT)
- Does not show the *reason* for delays or variances
- Limited in showing resource loading or conflicts
- Static once printed; requires frequent updating to remain useful
- Can create a false sense of certainty — bars look precise even when estimates are rough

- Does not capture the logic of why tasks must occur in a particular order

B20: Lead Time and Lag Time

These are scheduling concepts that describe the relationship between dependent tasks in a project network.

Lead Time: An *acceleration* of the successor activity relative to the predecessor — the successor starts *before* the predecessor has fully finished. Lead time is expressed as a negative value.

Example: In a construction project, interior fitting (successor) begins 2 weeks before the exterior cladding (predecessor) is fully completed, because the lower floors are ready while upper floors are still being clad. Lead = -2 weeks.

Lag Time: A *delay* inserted between the predecessor finishing and the successor starting — a mandatory waiting period. Lag time is expressed as a positive value.

Example: After pouring concrete (predecessor), a 28-day curing period must pass before structural loading (successor) can begin. Lag = +28 days.

Why they matter: Lead and lag times allow schedulers to model real-world sequencing constraints more accurately than simple Finish-to-Start relationships. Incorrect lead/lag assumptions are a common source of schedule error and can affect the critical path calculation.

B21: Critical Path vs Critical Chain

	Critical Path Method (CPM)	Critical Chain Method (CCM)
Focus	Task durations and logical dependencies	Resource constraints and human behaviour (padding)
What it identifies	The longest sequence of dependent tasks — determines minimum project duration	The longest chain of tasks accounting for both dependencies <i>and</i> resource availability
Buffer approach	No explicit buffers built in — safety time is embedded in individual task estimates	Strips individual safety time from estimates; consolidates buffers at the end of chains (project buffer) and before critical feeding tasks (feeding buffers)
Addresses	Schedule logic	Student syndrome (starting late) and Parkinson's Law (work expands to fill time)
Advantage	Simple, widely understood; identifies tasks that cannot be delayed	More realistic; prevents multi-tasking and hoarding of float
Limitation	Ignores resource constraints; safety padding in individual tasks inflates overall duration	More complex to implement; requires cultural shift in how estimates are provided

B22: Critical Path — Identification from a PERT Network

Definition: The critical path is the longest sequence of logically connected activities in the project network. It determines the **minimum project duration** — any delay on a critical path activity directly delays the project end date. Critical path activities have **zero float (slack)**.

How to identify it:

- 1. Draw a PERT network diagram with all activities, durations, and dependencies.
- 2. **Forward pass** — calculate the Earliest Start (ES) and Earliest Finish (EF) for each activity: $EF = ES + \text{Duration}$.
- 3. **Backward pass** — calculate the Latest Finish (LF) and Latest Start (LS) from the project end: $LS = LF - \text{Duration}$.
- 4. **Calculate float** for each activity: $\text{Float} = LS - ES$ (or $LF - EF$).
- 5. **Critical path** = the path where float = 0 on all activities.

Example:

Network:
A(3) → C(4) → E(2) Path 1: 3+4+2 = 9 days
A(3) → B(5) → D(1) Path 2: 3+5+1 = 9 days ← both paths critical (no float)
 ↘ F(6) Path 3: 3+5+6 = 14 days ← CRITICAL PATH

Critical path: A → B → F = 14 days
Float on path C-E: 14 - 9 = 5 days float (not critical)

Practical insight: Knowing the critical path allows the project manager to focus monitoring and control resources on activities that will directly impact the project end date, and to identify where "crashing" (adding resources to shorten duration) will be most effective.

B23: Risk Register — Purpose and Structure

Purpose: A Risk Register is a centralised document used to identify, record, assess, prioritise, assign, and track all project risks throughout the project lifecycle. It is the primary tool for implementing a systematic risk management process.

Typical structure — columns in a Risk Register:

Column	Content
Risk ID	Unique identifier for tracking
Risk Description	What might happen and why
Category	Type of risk (technical, schedule, cost, stakeholder, environmental, etc.)
Probability	Likelihood of occurrence (1–5 scale, or Low/Medium/High)
Impact	Consequence if it occurs (1–5 scale, or Low/Medium/High)

Column	Content
Risk Score	Probability × Impact (priority score)
Risk Owner	Person responsible for monitoring and responding to the risk
Response Strategy	Accept / Mitigate / Transfer / Avoid / Enhance / Exploit / Share
Response Actions	Specific actions to be taken
Residual Risk	Risk remaining after response actions
Status	Open / In Progress / Closed

Why it matters: The Risk Register transforms risk management from a one-time brainstorm into an ongoing, accountable process. It ensures risks are not forgotten, that ownership is clear, and that response strategies are documented and reviewed at each project stage.

B24: Risk Matrix for Prioritising Project Risks

A **Risk Matrix** (also called a Probability–Impact Matrix or 5×5 Priority Grid) plots identified risks against two axes — **Probability** (likelihood of occurrence) and **Impact** (severity of consequence) — to produce a priority score.

		IMPACT					
		1	2	3	4	5	
PROB	5	5	10	15	20	25	← Senior mgmt
	4	4	8	12	16	20	← Senior mgmt
	3	3	6	9	12	15	← PM intervention
	2	2	4	6	8	10	← PM intervention
	1	1	2	3	4	5	← Existing controls

Intervention thresholds (WK6a):

- Score **1–6**: Manage with existing controls; monitor.
- Score **8–12**: Project Manager intervention required.
- Score **15–25**: Senior management / escalation required.

How it is used: Each identified risk in the Risk Register is scored (Probability × Impact). The matrix makes the priority order visual and defensible. High-scoring risks receive immediate response strategies; low-scoring risks are monitored. The matrix is reviewed and updated at each project stage as new risks emerge or existing risks change.

B25: Positive Risks (Opportunities) vs Negative Risks (Threats)

Definition of risk (WK6a): Risk = the possibility of loss or injury; a situation with a measured impact — **positive or negative** — on a project outcome.

	Negative Risk (Threat)	Positive Risk (Opportunity)
Nature	An uncertain event that would harm project objectives	An uncertain event that could benefit project objectives
Impact direction	Increases cost, delays schedule, reduces quality	Reduces cost, accelerates schedule, improves quality
Example	A key supplier goes bankrupt mid-project	A new technology becomes available that halves installation time
Response strategies	Avoid, Mitigate, Transfer, Accept	Exploit, Enhance, Share, Accept
Common mistake	Treating all risk as negative — missing opportunities	Failing to actively pursue opportunities

Key insight: Project managers who focus only on threats miss opportunities to improve project outcomes. A mature risk management framework (e.g. ISO 31000) explicitly addresses both.

B26: Risk Response Strategies

(WK6a — for both threats and opportunities)

For Negative Risks (Threats):

Strategy	Definition	Example
Avoid / Reject	Eliminate the risk entirely by changing plans	Change project route to avoid an unstable geological zone
Mitigate	Reduce probability and/or impact to an acceptable level	Install surge protectors to reduce risk of equipment damage
Transfer	Shift the financial consequence to a third party	Take out insurance; use a fixed-price contract
Accept	Acknowledge the risk and take no proactive action (active: set contingency; passive: deal with it if it happens)	Accept the risk of minor weather delays with a contingency reserve

For Positive Risks (Opportunities):

Strategy	Definition	Example
Exploit	Ensure the opportunity definitely occurs	Assign your best team to a task that could finish early, guaranteeing the early completion
Enhance	Increase probability or impact of the opportunity	Add resources to accelerate a task that, if completed early, unlocks follow-on activities

Strategy	Definition	Example
Share	Allocate ownership of the opportunity to a third party better positioned to capture it	Partner with a specialist firm that can leverage the opportunity more effectively
Accept	Be willing to take advantage if the opportunity arises, but don't actively pursue it	Note that a favourable exchange rate might reduce material costs — take advantage if it occurs

B27: Earned Value Management (EVM) — Metrics, Formulas, and Interpretation

(WK4a)

EVM provides an integrated view of project scope, schedule, and cost performance at any point in time.

Core metrics:

Metric	Definition	Formula
Planned Value (PV)	The budgeted value of work <i>scheduled</i> to be done by a given date (also called BCWS — Budgeted Cost of Work Scheduled)	$PV = \text{Planned \% complete} \times \text{Budget at Completion (BAC)}$
Earned Value (EV)	The budgeted value of work <i>actually completed</i> by that date (BCWP — Budgeted Cost of Work Performed)	$EV = \text{Actual \% complete} \times \text{BAC}$
Actual Cost (AC)	The actual cost incurred for the work completed to date (ACWP)	Taken from financial records
Schedule Variance (SV)	Difference between EV and PV — is the project ahead or behind schedule?	$SV = EV - PV$
Cost Variance (CV)	Difference between EV and AC — is the project over or under budget?	$CV = EV - AC$

Interpretation:

- $SV > 0$ → Ahead of schedule. $SV < 0$ → Behind schedule. $SV = 0$ → On schedule.
- $CV > 0$ → Under budget (good). $CV < 0$ → Over budget. $CV = 0$ → On budget.

Additional indices:

- **Schedule Performance Index (SPI)** = EV / PV . $SPI > 1$ = ahead; $SPI < 1$ = behind.
- **Cost Performance Index (CPI)** = EV / AC . $CPI > 1$ = under budget; $CPI < 1$ = over budget.
- **Estimate at Completion (EAC)** = BAC / CPI (forecast of total project cost).

Example: BAC = \$100,000. At month 6, work is 40% complete (EV = \$40,000) but \$50,000 has been spent (AC = \$50,000) and 50% should have been done (PV = \$50,000).

- $SV = \$40,000 - \$50,000 = -\$10,000$ (behind schedule)
- $CV = \$40,000 - \$50,000 = -\$10,000$ (over budget)

- $CPI = 40,000/50,000 = 0.8 \rightarrow$ For every \$1 spent, only \$0.80 of value is delivered.

B28: EVM Advantages and Limitations

Advantages:

- Provides an **integrated view** of scope, schedule, and cost in a single set of metrics
- Enables **early warning** — problems are visible before they compound into major overruns
- **Objective** — based on work actually completed, not self-reported percentage progress
- Supports **forecasting** — EAC projects the final cost based on current performance trends
- Creates accountability — all team members know their work has a measurable value
- Widely accepted and required on many government and defence contracts

Limitations:

- Requires accurate and timely data — garbage in, garbage out
- The definition of "**% complete**" is subjective and open to manipulation ("80% syndrome")
- Can be complex and time-consuming to implement, especially for small projects
- Does not capture **quality** of work delivered — a task can be "complete" but substandard
- Focuses on cost and schedule; does not directly address stakeholder satisfaction, risk, or TBL
- Not well-suited to early project phases when scope is still being defined

B29: Key Elements of Quality Management in Projects

Quality management ensures the project satisfies the needs — standards, definitions, specifications, and intended use — for which it was undertaken. **(WK4b)**

The four stages ("The road to quality"):

1. **Quality Planning** — All mandated quality standards, operational definitions, and business requirements are clearly identified, agreed, and confirmed to be achievable and measurable. *Key principle: quality is planned in from the start; not randomly inspected in later.*
2. **Quality Assurance (QA)** — A guarantee that overall project performance (input, process, and output) is evaluated on a regular basis to give stakeholders confidence that quality standards will be satisfied. Tools: quality management plan, regular audits, ISO standards (ISO 9000 for quality, ISO 14000 for environmental management, ISO 26000 for social responsibility).
3. **Quality Control (QC)** — Monitors specific task and project results to identify, measure, and eliminate causes of unsatisfactory performance. *Control implies measurement.* Tools: peer reviews, physical inspection, control charts, scatter diagrams, checklists, Pareto diagrams, statistical sampling, flowcharts, fishbone/cause-and-effect diagrams, trend analysis.
4. **Quality (Continuous) Improvement** — A culture and commitment to *what* is being delivered and *how well*. Innovation, risk-taking, and efficiencies support this. Tools: regular performance reporting, decision gates, scenario analysis, evaluation reports, suggestion boxes, user feedback.

B30: Quality Management and the Triple Bottom Line (TBL)

Quality management and TBL are mutually reinforcing frameworks. Both require a project to deliver outcomes that are **fit for purpose across economic, social, and environmental dimensions**.

- **Economic quality:** Delivering on time and within budget is itself a quality outcome. The Balanced Scorecard's Financial Perspective captures this. Total Cost of Quality (see B33) shows that investing in prevention and appraisal reduces the far higher costs of failure — an economic quality argument.
- **Social quality:** Meeting the needs of the communities affected by the project is a social quality obligation. Quality standards that incorporate accessibility, safety, and cultural sensitivity directly address the "People" dimension of TBL.
- **Environmental quality:** ISO 14001 (Environmental Management) is a quality standard. Applying environmental quality controls — minimising waste, protecting habitats, managing emissions — directly addresses the "Planet" dimension. ISCA ratings for infrastructure projects are a direct intersection of quality assurance and environmental TBL.

Key link: The finalisation stage requires TBL review as a quality assurance mechanism — the project is not "complete" until all three dimensions of its stated purpose have been evaluated. Quality checklists are a useful tool to ensure TBL is considered during quality analysis (WK4b).

B31: ISO Standards in Project Quality Management

ISO (International Organisation for Standardisation) provides internationally recognised quality management standards that give projects a credible, auditable framework for quality assurance.

Key standards relevant to project management:

- **ISO 9000 series (Quality Management Systems):** Establishes principles for consistent quality delivery — customer focus, leadership, engagement of people, process approach, improvement, evidence-based decision making, relationship management. Provides the framework for a formal Quality Management Plan.
- **ISO 14000 series (Environmental Management Systems):** Provides a framework for controlling environmental impacts. ISO 14001 is the most widely used and requires organisations to establish environmental policies, identify impacts, set targets, and conduct audits. Directly relevant to the Planet dimension of TBL.
- **ISO 26000 (Social Responsibility):** Guidance on social responsibility across organisations. Addresses community engagement, labour practices, human rights, and stakeholder involvement. Supports the People dimension of TBL.
- **ISO 31000 (Risk Management):** Provides a framework and principles for risk management applicable to any organisation or project. Supports the risk management stage of project quality assurance.

Role in projects: ISO standards provide external benchmarks against which project performance can be measured and audited. They support supplier qualification, demonstrate due diligence to clients and regulators, and build confidence in quality outcomes.

B32: Benchmarking, Checklists, and Cause-and-Effect Diagrams as Quality Assurance Tools

(WK4b — Quality Assurance tools)

Benchmarking: Comparing a project's processes, costs, schedule performance, or quality outcomes against industry best practices or comparable projects. *Example: comparing construction waste generation per m² against industry benchmarks for sustainable construction.* Benchmarking identifies performance gaps and provides targets for improvement. Limitation: benchmarks may be outdated or drawn from non-comparable projects.

Checklists: Standardised lists of items that must be verified or completed at a specific stage of the project. Used in quality control to ensure nothing is missed. *Example: a pre-pour concrete checklist verifying rebar placement, formwork, and mix design before pouring.* Advantages: simple, fast, systematic, reduces reliance on memory. Limitation: checklists can promote checkbox mentality — items are ticked without genuine verification.

Cause-and-Effect Diagrams (Fishbone / Ishikawa Diagrams): A structured visual tool for tracing a quality problem back to its root causes. The "effect" (defect or problem) is placed at the right; possible causes are organised into branches (commonly: Materials, Methods, Machines, People, Environment, Measurement). *Example: for a recurring delay in report delivery — causes might be identified across People (skill gaps), Methods (unclear process), or Materials (incomplete data).* Helps teams avoid fixing symptoms rather than causes.

B33: Total Cost of Quality (TCoQ)

Definition: Total Cost of Quality is the sum of all costs associated with ensuring products and services meet quality standards — including both the costs of achieving quality and the costs of failing to achieve it.

Four components:

Category	Type	Description	Examples
Prevention Costs	Cost of good quality	Investment to prevent defects from occurring	Training, quality planning, design reviews, process improvements, supplier qualification
Appraisal Costs	Cost of good quality	Testing and inspection to detect defects before delivery	Quality audits, testing, inspections, peer reviews
Internal Failure Costs	Cost of poor quality	Costs of defects found <i>before</i> delivery to the client	Rework, scrap, re-testing, design changes
External Failure Costs	Cost of poor quality	Costs of defects found <i>after</i> delivery	Warranty claims, legal action, reputation damage, re-work at client site

Key insight: Prevention and appraisal costs are investments that reduce the far larger internal and external failure costs. The optimal quality strategy invests heavily in prevention and appraisal upfront — not because it is "nice to have" but because it is economically rational. External failure costs (e.g. legal action, reputational damage) can be catastrophic and exponentially exceed prevention costs.

B34: Communication Barriers and How to Overcome Them

(WK5b)**Common barriers in projects:**

- Lack of client involvement — client disengages; requirements drift
- Poorly informed stakeholders — decisions made on incomplete information
- Lack of meetings, or too many meetings with no action
- Lack of reporting requirements — no formal obligation to communicate
- Poor and incomplete documentation
- Frequent scope change — disrupts communication channels and understanding
- Changing project personnel — institutional knowledge is lost
- Lack of auditing to identify lessons learned
- Cognitive / perceptual differences — the same message is decoded differently by different receivers (Belch & Belch 2004 — sender → encoding → message → channel → decoding → receiver, with *noise* disrupting at any step)

How to overcome them:

- Establish a formal **Communication Management Plan** — defining who receives what, when, in what format, and through which channel
- Use the right communication medium for each audience (formal reports for executives; team meetings for technical issues; newsletters for community stakeholders)
- Separate **Fact** from **Fantasy / Folklore / Feelings** in all project communications
- Conduct regular, structured stakeholder engagement — don't wait for problems to surface
- Create feedback loops — acknowledge receipt; confirm understanding
- Document all key communications and decisions
- Manage personnel transitions carefully — formal knowledge transfer procedures

B35: Project Meeting Tools — Agendas, Minutes, and Stakeholder Profiling

Agendas: A structured list of topics to be covered in a meeting, distributed in advance. Benefits: ensures meetings are purposeful and focused; allows participants to prepare; prevents agenda capture by dominant voices; establishes expected duration. A good agenda includes: topic, presenter/owner, time allocation, and desired outcome (decision vs discussion).

Meeting Minutes: A written record of what was discussed, decided, and assigned during a meeting. Key components: attendees; topics discussed; decisions made; action items (with owner and due date); next meeting date. Minutes serve as the official record of commitments and are used to follow up on actions at the next meeting. Without minutes, decisions are forgotten and accountability is lost.

Stakeholder Profiling Techniques: Tools used to understand stakeholder characteristics before determining how to engage with them:

- **Power–Interest Matrix** (Mendelow) — see B5
- **RACI / PARIS matrices** — see B6
- **Stakeholder register** — records name, role, interests, influence, communication preferences, and current attitude (supporter/neutral/opponent)

- **Salience Model** — classifies stakeholders by Power, Legitimacy, and Urgency (Mitchell et al.)

Together, profiling techniques ensure communication is not generic but tailored — the right message, in the right format, at the right frequency, to the right stakeholder.

B36: Project Performance Reports — Progress Reports, Status Reports, and Forecasts

Performance reporting keeps stakeholders informed of project progress and supports decision-making.
(WK5b / WK7)

Progress Reports: Focus on what has been *accomplished* during a specific reporting period. Typically prepared by the project manager and submitted to the sponsor/steering committee. Content: activities completed, milestones achieved, work packages in progress, deliverables submitted. Frequency: weekly or fortnightly during active execution.

Status Reports: A snapshot of the project's current condition relative to the approved baseline at a specific point in time. Content: summary of schedule performance (ahead/on/behind), cost performance (under/on/over budget), scope status, risk status, issues log, action items. Often uses a traffic-light system (green/amber/red) for quick executive scanning. Status reports typically incorporate EVM data (SV, CV, SPI, CPI) for quantitative performance assessment.

Forecasts: Forward-looking — project future project performance based on current trends. Key forecasts: Estimate at Completion ($EAC = BAC / CPI$), Estimate to Complete ($ETC = EAC - AC$), projected milestone dates. Forecasts allow proactive decision-making — if a project is trending 10% over budget, corrective action can be taken now rather than at project end.

Best practice: All three types of report should be provided to appropriate stakeholders on a defined schedule, using consistent formats, so that trends are visible over time. They should be accurate, timely, and actionable — not just a compliance exercise.

SECTION C — Ethics, Leadership & Sustainability (Flagged Topic Questions)

C1: Morality, Moral Philosophy, and Ethical Decision-Making

Morality: The set of values and principles that individuals or groups hold about what is right and wrong. Morality is often culturally embedded and personal — it is the *lived* system of values by which a person acts. Example: a belief that it is wrong to bribe a contractor, even if it would accelerate the project.

Moral Philosophy (Ethics as a discipline): The systematic, theoretical study of morality — what it is, why it exists, and how moral questions should be reasoned about. Moral philosophy produces ethical *theories* (utilitarianism, deontology, virtue ethics) that provide frameworks for evaluating whether actions are right or wrong. It moves from personal morality to rational, universal principles.

Ethical Decision-Making: The process of applying moral philosophy and personal morality to a real-world dilemma in order to choose a course of action. It involves: recognising an ethical issue exists; gathering facts; applying ethical frameworks; considering stakeholder impacts; making a decision; and reflecting on outcomes (consistent with the 7-stage Ethical Cycle — see C7 reference below).

Distinction: Morality is *what you believe*. Moral philosophy is *how you reason about it systematically*. Ethical decision-making is *how you act in practice*, drawing on both. A project manager may personally believe bribery is wrong (morality), understand why it is wrong through Kant's categorical imperative (moral philosophy), and refuse a corrupt payment under pressure (ethical decision-making).

C2: Moral Relativism, Humanism, and Egoism Within Virtue Ethics

These are three sub-positions within or related to virtue ethics that explain different ways individuals approach the question of what is "good."

Moral Relativism: The view that moral standards are not universal but relative to culture, time, or individual perspective. What is considered virtuous or ethical in one culture may not be in another. *Example: gift-giving norms that would be considered bribery in Australia may be standard business courtesy in another country.* Moral relativism challenges universal codes of ethics but risks becoming an excuse for harmful practices ("it's just their culture"). Project managers working internationally must navigate the tension between moral relativism and universal ethical standards.

Humanism: The view that human beings are the primary source of moral value, and that human flourishing (eudaimonia — Aristotle's concept) is the goal of ethical life. Virtue ethics is inherently humanist: it focuses on what kind of person one should be and what human capacities (courage, honesty, prudence, justice) should be cultivated. In projects, humanism manifests as a commitment to treating all stakeholders — workers, communities, future generations — as ends in themselves, not merely as means to project delivery.

Egoism: The view that an individual's primary moral obligation is to act in their own self-interest. Ethical egoism argues that each person *ought* to do what is best for themselves; psychological egoism argues that people *do* in fact act from self-interest regardless of stated motives. In project management, egoism is a warning: a project manager who acts primarily from self-interest (career advancement, bonus targets) rather than stakeholder good will make systematically poor ethical decisions. Kant's categorical imperative directly counters egoism — if everyone acted purely in self-interest, cooperation and trust would collapse.

C3: Legal Compliance vs Professional Ethics — Why Engineers Must Satisfy Both

Legal compliance means adhering to the laws, regulations, and statutory requirements that govern engineering practice. The law sets a *minimum standard* — what is legally required.

Professional ethics means adhering to the higher standards of conduct embodied in professional codes of ethics (e.g. Engineers Australia Code of Ethics) — standards that reflect the profession's fiduciary relationship with the public.

Why both are necessary:

1. **The law is a floor, not a ceiling.** A project manager can comply fully with all legal requirements and still act unethically. *Example: a mining company legally permitted to discharge waste into a river may still cause environmental harm that, while lawful, violates the ethical principle of "promote sustainability."*
 2. **Codes of ethics fill legal gaps.** Laws cannot anticipate every situation. Professional codes provide guidance in novel or ambiguous circumstances where legislation is silent.
 3. **Public trust requires more than legal compliance.** Engineers operate in a position of **fiduciary trust** with the public — the public lacks the technical expertise to assess engineering decisions themselves. This trust demands a higher standard than "we didn't break any laws."
 4. **Engineers Australia's position:** Members must demonstrate integrity, practice competently, exercise leadership, and promote sustainability — all of which go beyond statutory minimum requirements.
 5. **Professional consequences:** A member can be legally compliant but still face professional sanctions (suspension, expulsion) for violating the code of ethics.
-

C4: Engineers Australia Code of Ethics — Four Principles

(WK10b)

1. **Demonstrate Integrity** Act with honesty and transparency in all professional conduct. This includes being truthful in reports, disclosing conflicts of interest, maintaining confidentiality, and refusing corrupt practices. *Example: disclosing that a recommended supplier is also a personal associate, even if it risks the business relationship.*
2. **Practice Competently** Only accept work within your area of competence; maintain and develop skills; seek assistance when needed; take responsibility for your work. Engineering decisions can affect public safety — incompetent practice is not merely an ethical failing but a public safety risk. *Example: a civil engineer who is not qualified in geotechnical engineering should not certify a foundation design without appropriate expert input.*
3. **Exercise Leadership** Actively promote the adoption of ethical and sustainable practices within your team, organisation, and profession. Leadership here is not merely managerial authority — it is the ethical obligation to speak up, challenge unethical decisions, and model the behaviour you expect from others. *Example: a project engineer who raises concerns about a cost-cutting measure that compromises structural safety, even under commercial pressure.*
4. **Promote Sustainability** Balance economic, social, and environmental considerations in all professional decisions — consistent with the TBL framework. Consider impacts on current and future generations (life-cycle thinking). *Example: specifying recycled materials for a construction project where they meet technical requirements, even if virgin materials are marginally cheaper.*

Underlying moral commitments: Honesty and integrity, impartiality, fairness and equity, and dedication to the protection of public health, safety, and welfare.

C5: Compromise and Integrity in an International Project Environment

Compromise in an international project environment refers to adjusting one's approach to accommodate different cultural, legal, and business norms — while *not* abandoning core ethical principles.

When compromise is acceptable:

- Adapting communication styles to cultural norms (e.g. more formal or hierarchical communication in some contexts)
- Adjusting contract structures to reflect local legal frameworks
- Adapting scheduling to local religious observances or public holidays
- Accepting different procurement practices that are legal and transparent locally

Example: In some countries, presenting a formal gift at the commencement of a project is a cultural norm, not a bribe. A project manager can accept this practice while maintaining clear policies against gifts that create actual conflicts of interest or obligations.

When compromise is not acceptable (maintaining integrity):

- Falsifying safety records to satisfy local regulators who expect lower standards
- Tolerating child labour because it is common locally
- Paying facilitation payments ("speed money") to government officials
- Misrepresenting project costs in reports to satisfy host-country partners

How to navigate the tension:

- Apply Schwartz's 6 universal moral values (trustworthiness, respect, responsibility, fairness, caring, citizenship) as a baseline that transcends cultural difference.
- Use the Ethical Cycle (7 stages) to work through dilemmas systematically.
- Consult the UN Global Compact (10 principles) as an internationally recognised ethical standard.
- The **line-drawing approach**: place the proposed compromise between a clear positive paradigm (unambiguously ethical) and a clear negative paradigm (unambiguously unethical). If the action sits clearly closer to the negative paradigm, it must be refused.

C6: Balancing Host-Country Practices with Universal Ethical Standards

A multinational project manager faces four options when home-country and host-country ethical norms conflict (Beauchamp et al. 2008, WK11b):

1. **Follow home-country norms** — Apply the standards of your own country regardless of local context. Risk: cultural insensitivity, commercial disadvantage, or practical impossibility.
2. **Follow host-country norms** ("When in Rome...") — Adopt local standards entirely. Risk: ethical relativism; may require practices that violate universal standards (bribery, discrimination, unsafe working conditions).
3. **Follow whichever norm is most profitable** — Purely consequentialist approach. Risk: highly unethical in practice; likely to produce harmful outcomes when short-term profit and ethics diverge.
4. **Follow whichever norm is morally best** — The recommended approach: apply the higher of the two standards, or the standard most consistent with universal moral values. This may require courage and commercial sacrifice.

Practical strategies for project managers:

- Establish a clear ethical framework *before* entering a new jurisdiction — agree on non-negotiables with leadership.
- Use the **6 universal moral values** (Schwartz 2005) as an anchor: trustworthiness, respect, responsibility, fairness, caring, citizenship. These transcend cultural variation.
- Apply the **9 ethical tests** (see C7) to any proposed action that feels uncomfortable.
- Build relationships with local ethical leaders who can advise on navigating cultural norms without compromising integrity.
- Remember: complying with local law is necessary but not sufficient — professional ethics standards apply globally.

C7: Five Ethical Tests for Project Managers

(WK10a — full list contains 9 tests; 5 commonly examined)

A project manager can apply the following tests to determine whether an action is ethical (Test → Explanation → Example):

1. Legal Test *Does the action violate the law?* The law represents the minimum ethical standard. If an action is illegal, it fails the most basic ethical test regardless of other justifications. *Example:* A project manager is asked to approve a contractor's invoice that includes fictitious items to cover a bribe payment. This violates Australian anti-bribery law (Criminal Code Act 1995) — it fails the Legal test immediately.

2. Consequences Test *Will the action result in major harm to people or organisations?* Drawn from utilitarian thinking — the action is assessed by the harm it might cause, even if it is technically legal. *Example:* Cutting the thickness of load-bearing walls to reduce costs is legal if it technically meets minimum code. But if this materially increases collapse risk, it fails the Consequences test.

3. Rights Test *Does the action infringe upon the rights of stakeholders?* Every stakeholder has rights — to safety, to honest information, to be treated with dignity. An action that violates those rights is ethically unacceptable even if no law is broken. *Example:* Disposing of construction waste near a community water source without disclosing this to the community infringes their right to information and to a safe environment.

4. Justice Test *Does the action leave another person or group less well off?* Based on distributive justice — fair allocation of benefits and burdens. If one group bears disproportionate costs of a project without corresponding benefit, this test is failed. *Example:* A road upgrade generates significant economic benefit for commercial districts but routes noise and heavy traffic through a low-income residential neighbourhood exclusively — the justice test questions whether this distribution is fair.

5. Sustainability Test *Does the action negatively impact future generations?* Directly aligned with TBL and life-cycle thinking. Any action that compromises the ability of future people to meet their own needs fails this test. *Example:* A project that permanently destroys coastal habitat to create a marina precinct may pass economic tests in the short term but fails the Sustainability test if the habitat loss is irreversible and its ecological value to future generations is unaccounted for.

Appendix: 7 Stages of the Ethical Cycle (Van de Poel & Royakkers 2007, WK10b)

1. **Case** — The cycle starts with a particular case of moral complication.

2. **Moral problem statement** — Precise formulation: what is the problem, for whom, and what is its moral nature?
 3. **Problem analysis** — Identify: (a) stakeholders; (b) moral values involved; (c) relevant facts.
 4. **Options for action** — Identify all possible options — *thinking creatively is essential*.
 5. **Ethical judgement** — Apply moral concepts, ethical theories, models, and frameworks (utilitarian, deontological, virtue ethics, codes of ethics).
 6. **Reflection** — Does the chosen action stand up to scrutiny? Consider objections.
 7. **Morally accepted action(s)** — Implement the chosen action to resolve the dilemma.
-

Document prepared for GENG5505 exam revision | Sources: section a answer.md (ORM case), 5505_corrected.md (lecture-verified content), lecture slides WK1–WK12